CHAPTER

23

Finishing and Polishing

The main objective of restorative dentistry is to restore the individual tooth to its form and functions along with imparting pleasing esthetics and maintaining periodontal tissues in good esteem. Many a times, the form and function is restored but the surface may not be smooth. It has been established that rough/uneven surfaces invite microbial flora; also the light reflected by these surfaces is not uniform. A restored tooth should be evenly smooth and reflect light uniformly. The process of making the surface smooth, reflecting the light evenly, is described as *finishing and polishing*.

DEFINITIONS

Finishing is the removal of surface irregularities; whereas, polishing is the creation of surface layer, which can reflect light as good as enamel surface.

'*Finishing*' is defined as "*the transformation of an object from a rough to a refined form. The procedure involves removal of surface irregularities and shaping the restoration according to functional occlusion*". '*Polishing*' is defined as "*the production of a shiny, mirror like surface, which reflects light similar to enamel without creating supplemental films by the addition of wax or lacquer*".

'Buffing' refers to the polishing process in which the abrasives are applied via bristle brushes (for leather and cloth materials).

'Finishing' can be described as a process whereby substrate particles are removed by the action of cutting and/or grinding. The surface of the substrate is made to come into frictional contact with a comparatively harder material. This contact generates enough tensile and shear stresses to overcome the forces of atomic bonds and thereby release particles from the substrate. In a 'cutting operation', the substrate particles are removed by the use of any instrument, which acts in a blade-like fashion. A 'grinding operation' on the other hand removes small particles of the substrate through the action of bonded or coated abrasive instruments.

The process of finishing leads to:

- Gross reduction of excess restorative material
- Contouring: Reproduction of size, shape, pits/grooves and other details of the tooth form.
- Establishes a well-adapted tooth–restoration junction
- Removes gross scratches.

'Polishing' is the most refined process and acts on an extremely thin region of the substrate surface. It produces very fine scratches that can be visible only under high magnification. An ideally polished surface does not have any surface imperfections, but this condition is very difficult to achieve because most of the restorative materials are brittle and get damaged even during fine grinding.

The polishing material does not cut or grind, but fills fine scratches and produces a perfectly smooth surface. During the polishing of metals, a highly microcrystalline layer is formed on the surface, called the '*Beilby layer*'. This layer is composed of non-oriented crystals and is 20–40 Å thick. It is hypothesized that because of rapid movement of the polishing agent, top layer of the material gets heated up, causing it to flow and fill the scratches.

The optimum speed for polishing varies with the hardness of material. The harder the material, higher should be the speed (Chrome cobalt alloys require higher speed than gold alloys).

The final surface character achieved on a restorative material is the combination of finishing and polishing. The polished surface can also be obtained by a coating, viz. electroplated deposits, sealants, unfilled resin and overglazes on porcelain.

The following factors influence the effectiveness of finishing/polishing procedures:

- Physical/mechanical properties of the restorative material
- Hardness, size and shape of the abrasive used
- Difference of hardness between the restorative material and the abrasive used. (Greater the difference, more will be the abrasive effect; for example, hardness of porcelain (Moh's scale) is 7 and the hardness of tungsten carbide/aluminum

oxide is 9 : the latter can effectively finish porcelain surface)]
- Speed and pressure at which abrasive is applied.
- Judicious use of lubricants (petroleum jelly, glycerol, silicone grease, etc.) along with abrasives.

Objectives of Finishing and Polishing

The objectives of finishing and polishing are:
- Adequate adaptation and continuity of the restoration margins with the tooth surface.
- Optimum restoration contour; proper occlusal contacts.
- Surface free of any scratches or irregularities; biologically acceptable.
- Adjustment of minor morphological or marginal defects.
- Enhanced oral functions.
- Obtain smooth surface, which can reflect light similar to that of enamel (*A surface becomes reflective only when surface irregularities are well below 10 µm; below the resolution of wavelength of light*).
- Improve oral hygiene.

Finishing and Polishing Instruments

a. ***Finishing burs:*** 12–30 fluted carbide burs are usually used for finishing depending upon the restorative material. Plain burs are also used before final polishing. Burs are available in various sizes and shapes, viz., straight fissure, tapering, flame shaped, rounded, wheel, etc. (Fig. 23.1).

b. ***Diamond instruments and pastes:*** Diamond is considered as a super abrasive because it is capable of abrading almost any other material. It is available in the form of bonded abrasive rotary instruments, metal backed abrasive strips and polishing pastes. Finishing diamonds of medium-fine grit contain diamond particles of 8.0–50.0 µm in diameter (Fig. 23.2). These instruments should always be used with a light speed (less than 50,000 rpm) and copious water spray to preserve the very fine diamond coatings. Diamond polishing pastes are also available with particle sizes ranging from 2.0–5.0 µm.

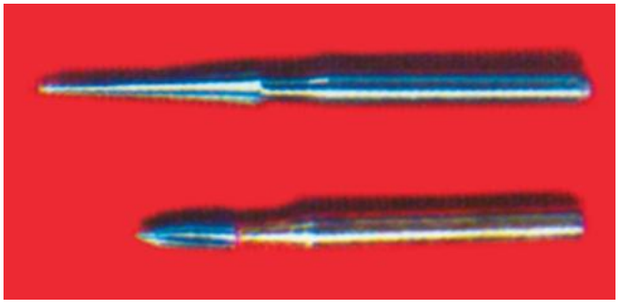


Fig. 23.1: Finishing burs



Fig. 23.2: Finishing diamonds

c. ***Brushes:*** Brushes can be used either alone or in combination with the abrasive particles. Most of the brushes have synthetic bristles; a few may have wire bristles for polishing cast restorations. Various types of brushes are available, which may be inserted into a handpiece or attached to a mandrel (Fig. 23.3).

d. ***Cloth:*** Cloth carried on a metal wheel may be used for final polishing, with or without a polishing medium.

e. ***Felt:*** Felt is used to attain lustre for the metallic restorations usually with a polishing agent. It is available in different shapes of wheels, cones and cylinders.

f. ***Rubber instruments:*** Rubber-end tools are commonly used for polishing procedures. These are available in various shapes of cups, wheels, cones, etc., and are commonly used with other abrasives or

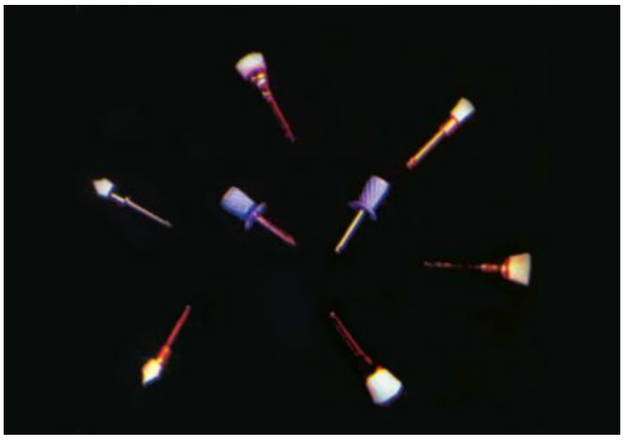


Fig. 23.3: Brushes and cups

polishing pastes (Fig. 23.3). These should not be used with heavy pressure, since excessive heat is produced. Most of these products contain latex (potential allergen) and often leaves a residue on the surface of the restoration.

g. ***Coated discs:*** The abrasives particles, such as sand, cuttle, garnet, boron carbide, silicon carbide, etc. are held onto a flexible heavy weight paper or mylar strips with a suitable adhesive material. The discs may be attached to a mandrel for rotary finishing (Fig. 23.4).

h. ***Strips:*** Abrasive strips are used by hand in a back and forth motion especially for finishing proximal areas (Fig. 23.5). They are available with metal or plastic backing with different abrasiveness. Metal strips can be autoclaved; whereas, plastic strips are for single use.

i. ***Stones:*** Stones are used for contouring and finishing, where maximum abrasion is needed, such as, adjusting the occlusion.

ABRASIVE MATERIALS

a. ***Chalk:*** It is a white-colored abrasive composed of calcium carbonate. It serves as a mild abrasive for polishing tooth enamel, gold foil, amalgam and plastic materials.

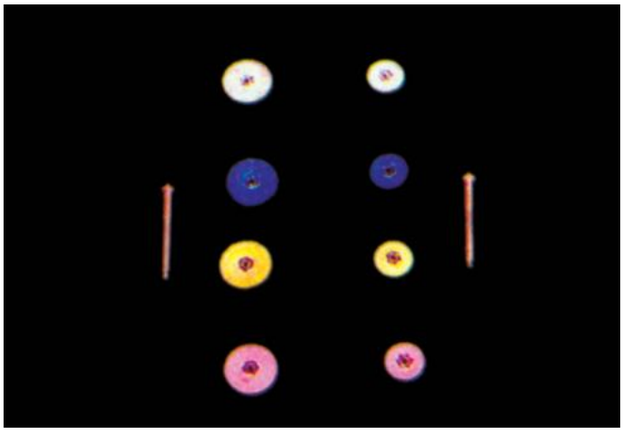


Fig. 23.4: Polishing discs

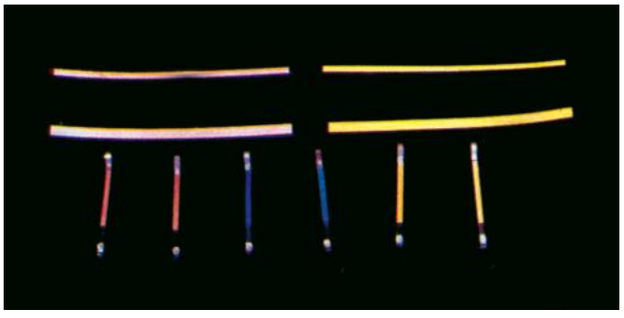


Fig. 23.5: Interproximal strips for finishing

b. ***Pumice:*** It is the most common abrasive, used mainly for softer materials. Pumice slurry is a mix of pumice and water whereas a pumice paste is a mix of pumice and glycerine. Pumice can be used for polishing tooth enamel, gold foil, amalgam and acrylic resins. Pumice powder is mixed with water or glycerine and carried to the surface by a revolving brush. Slow speed is recommended during its use; otherwise the centrifugal force throws out most of the pumice from the brush.

c. ***Corundum:*** This is a form of aluminum oxide (usually white) usually available in the form of bonded abrasive. Corundum is used primarily for grinding metal alloys.

d. ***Emery:*** Emery is greyish-black corundum; a mixture of aluminum oxide and iron. It is commonly used as an abrasive coating on a cloth or paper disc. It may be used for finishing metal alloys and plastic materials.

e. ***Garnet:*** Garnet includes a number of materials, such as silica, aluminum, cobalt, iron, magnesium and manganese. Its color is dark red and is extremely hard (highly abrasive). It is used in grinding metal alloys and plastic materials.

f. ***Sand:*** Sand is predominantly silica. Irregularly shaped particles are bonded to the paper with glue or resin. It is supplied in the form of sand paper. Sand particles under air pressure are also used to remove the investment material from base metal alloy castings.

g. ***Tripoli:*** The gray- and red-colored tripoli are commonly used for polishing metal alloys and some plastic materials.

h. ***Kieselguhr:*** In addition to its use as a fine abrasive on paper discs, it is also used as filler in various impression materials. The coarser form is called 'diatomaceous earth'. Chronic exposure to this material can cause respiratory silicosis.

i. ***Cuttle fish bone:*** It is available as coated discs and strips for polishing metal margins and amalgam restorations.

j. ***Quartz:*** Quartz is hard, colorless and transparent material. Sharp particles of quartz are coated over paper plastic discs. Quartz abrasives are mostly used to finish metal alloys.

k. ***Zirconium silicate:*** It is an off-white mineral used to make coated discs and strips. It forms an important component of the prophylaxis pastes.

l. ***Silicon carbide (carborundum):*** Mounted points, coated discs and grinding wheels are the most common form of carborundum. It has a green and blue-black color. Carborundum wears away rapidly at high heat; should be kept wet when in use. Boron carbide has also been used.

ADJUNCT FINISHING MODALITIES

Certain procedures, designed as adjust finishing modalities, are used to achieve smooth surfaces before the final process of finishing and polishing.

A. Microabrasion

The process of microabrasion is basically used to remove stained surfaces. A paste (18% hydrochloric acid, pumice and water) is applied to the desired surface with a slow driven rubber cup. The rubber cup should be rotated at very slow speed to avoid removing any excess tooth structure and to prevent spatter. If the defect is only confined to the superficial 0.2–0.3 mm of enamel, the acid pumice paste strips away the affected surface, resulting in improved appearance with clinically unrecognizable enamel loss. Acid causes dissolution of enamel and the abrasive causes removal of superficial stains and defects. After application of a topical fluoride solution and final finishing and polishing, normal enamel surface characteristics are established.

One such system, 'Air-flow handy system', uses sodium bicarbonate as an abrasive agent. The powder is passed through a narrow nozzle of specially designed handpiece with the help of compressed air (Fig. 23.6).

B. Burnishing

'*Burnishing*' is defined as "*the process of rubbing a metal over the restorative material to make it shiny or lustrous*". Further, it aids in adaptation and compaction of the material at the margins.

'Burnishing' has also been described as a procedure in which the surface of a metallic restoration is smoothened by rubbing it with another small, highly polished, hard metal surface.

Instrument tips used for burnishing are smooth faced and are available in various shapes, such as, bell shaped, egg shaped, apple shaped, etc. Special burnisher, like 'Spratley burnisher' is used for proximal gingival marginal burnishing of metallic restorations.

Burnishing is commonly employed for amalgam and gold foil restorations and is always carried out from the restoration surface to the tooth surface.

a. Burnishing of Amalgam Restorations

The appropriate time for burnishing amalgam restorations may be divided as the '*pre-carve burnishing*' and the '*post-carve burnishing*'. Pre-carve burnishing, as the name suggests, is carried out before the carving is initiated and is in fact a continuation of the condensation process. Large egg-shaped/ball-shaped burnisher is used with a pressure equal to that used for condensing amalgam. The burnisher is moved mesiodistally and faciolingually taking care that its head contacts only the cuspal slopes and not the margins. It is established that burnishing during condensation improves marginal adaptation both for the conventional and high copper amalgams.

'Post-carve burnishing' is carried out after the carving is completed. Instruments used for amalgam carving are shown in Fig. 23.7. A round burnisher of appropriate size is moved in light strokes from the amalgam to the tooth surface. Proximal portions can be similarly burnished using a beaver tail burnisher and/or Spratley burnisher. Caution should be exercised against using excessive pressure leading to heat generation. Temperature greater than 60°C can damage the pulp beyond repair and release mercury to the surface. The presence of mercury-rich layer at the margins promotes corrosion and may lead to fracture.The properties achieved by post-carve burnishing are: Increased surface hardness, decrease in the size and number of voids on the surface and marginal areas, a slower rate of corrosion, denser amalgam at the margins, and improvement in the marginal seal. With the high copper amalgams, post-



Fig. 23.6: Air flow polishing system



Fig. 23.7: Instruments used for carving amalgam restoration

carve burnishing can be avoided; no significant effect on their performance has been established.

b. *Burnishing of Direct Gold Restorations*

Spratley burnisher is used with considerable pressure across the metal surface to the tooth. The metal is pushed to close the voids and to ride over the enamel margins. The pressure and rubbing gives the gold its shine, even though the surface is not ideally smooth. For proximal surfaces, Spratley carver can be used in the embrasure area. Burnishing also increases the hardness of the metal surface by cold working.

c. *Burnishing of Cast Gold Restorations*

The time of burnishing a cast gold restoration varies with the type of cementing medium used. If zinc phosphate cement is used for cementing, burnishing is delayed and carried out after 24 hours. The burnishing is delayed so as to let the cement dissolve at the margins. Once the cement is dissolved, burnishing is carried out to achieve the gold–tooth interface rather than gold–cement–tooth interface. On the other hand, if glass-ionomer cement/resin cement is used, the burnishing can be carried out immediately after setting, since these cements are less soluble in oral fluids.

Hand burnisher and rotary burnisher may be employed for cast gold alloys.



Fig. 23.8: Rotary finishing stones



Fig. 23.9: Rotary finishing wheels

FINISHING AND POLISHING OF INDIVIDUAL RESTORATIONS

A. Amalgam Restorations

The cavity is slightly overfilled during amalgam condensation. The mercury-rich layer of the last increment is removed during carving. (Within 3–5 minutes, the amalgam is sufficiently hardened.) Clinically, a ringing sound is considered the ideal period for carving. It is opined that burnishing should be carried out both prior to and after carving to prepare the restoration for finishing and polishing. Burnishing produces ten-fold decrease in surface roughness.

The restoration is left undisturbed for a minimum period of 24 hours before it is finally finished and polished (time required for setting, hardening and dimensional changes of amalgam).

After 24 hours, the surface of the restoration is rough due to the heterogeneous structure of set amalgam.

Finishing is initiated using finishing burs/stones (Figs 23.8 and 23.9). Initial finishing help trimming excess amalgam at the margins; also correct occlusal disharmony, if any. A common error in the trimming of amalgam is that of leaving fins or flashes at the margins. Since amalgam has a low tensile strength, thin sections of amalgam can easily get fractured resulting in marginal ditching.

For the proximo-occlusal restorations, overhangs are removed by using periodontal files or gold foil knives. Finishing of cervical areas is carried out by moving the fine water-resistant strips cervical to the contact area through the interdental space. The facial and lingual proximal margins are finished with light cuttle fish sandpaper discs.

It is emphasized that abrasives should be applied in a descending grade, (i.e. coarse, medium, fine and ultra-fine), to achieve the final finish. The final polish/metallic lustre is achieved by the application of a suitable polishing agent, like tin oxide, zinc oxide, chalk, pumice, etc. carried with a soft rotating brush or rubber cup. For polishing in the gingival embrasures, the abrasive is introduced through polishing strips.

During polishing, the restoration should be kept moist; low rotational speeds with light intermittent pressure should be used to avoid any overheating. Excessive heat may damage the pulp and also tend to bring mercury to the surface.

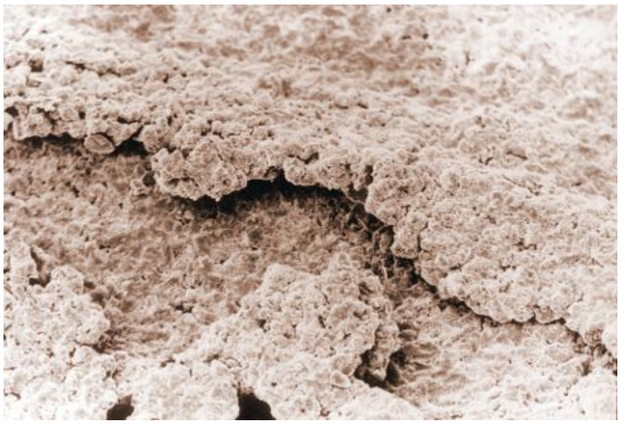


Fig. 23.10: Scanning electron microscopic view: Unpolished amalgam surface

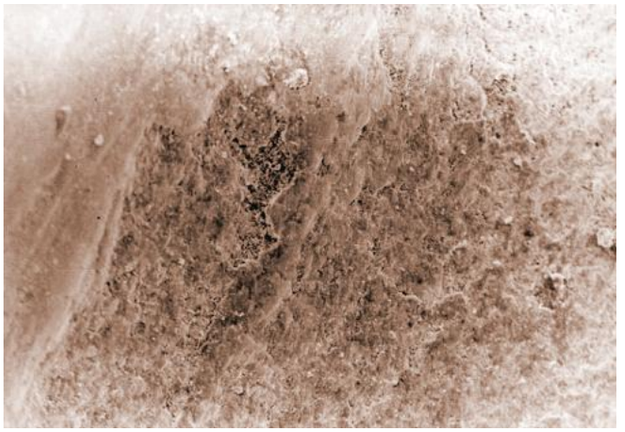


Fig. 23.11: Scanning electron microscopic view: Polished amalgam surface

High copper alloys can be polished 15–30 minutes after condensation. High copper amalgams have a rapid setting tendency and achieve high early compressive strength, allowing them to withstand the forces of finishing quite earlier.

The scanning electron microscopic picture of unpolished and polished amalgam is shown in Figs 23.10 and 23.11.

B. Composite Restorations

The features, which influence the process of finishing and polishing are as follows:

i. Conventional composites pose difficulty in achieving a smooth surface because of the difference in hardness of organic and inorganic phases. The resin matrix is soft and filler particles are hard. If fine grit polishing methods are used, the softer resin matrix abrades away easily, leaving behind the harder filler particles, which gives it a rough surface. If coarse abrasives are used, the organic and inorganic phases are removed equally, but the abrasive leaves behind rough marks. The unequal surfaces of conventional composites lead to unequal wear, which may further disturb the occlusal harmony.

ii. Hybrid composites can be polished to a semigloss state, but the surface is hydrophobic; unpleasant for the patient.

iii. Microfilled composites can be polished to the highest gloss state and are considered to be esthetically best amongst all composites.

A surface finish attained with the use of a plastic matrix band is the most desirable finish; however, this is rarely achieved, because of the need for contouring and removal of excess material. It is advisable to contour the unpolymerized composite with hand instruments, so that the need for removal of large amounts of set resin leading to surface damage are minimized.

The procedures for finishing and polishing of direct and indirect composite restorations are described separately:

a. *Direct Composite Restorations*

Excess composite at the cavosurface margins is scraped away using a scalpel or a sharp gold knife. The use of stainless steel instruments should be avoided, as these tend to leave gray marks on the restoration. The 30 fluted carbide burs and diamond points (15–45 μm size) are recommended for gross finishing (Fig. 23.12).

Rotary instruments should be used with absolute gentle pressure. High pressure tends to loosen and dislodge the filler particles. For the occlusal surface, cuttlefish or silicon dioxide-coated discs and strips are used in a descending grade of their abrasiveness. Finer discs are used for producing a smooth surface texture and refined marginal adaptation (Fig. 23.13). Vaseline or petroleum jelly should be used as lubricant with these discs and strips (Fig. 23.14A, B).



Fig. 23.12: Composite finishing kit



Fig. 23.13: Types of discs used for finishing and polishing composite

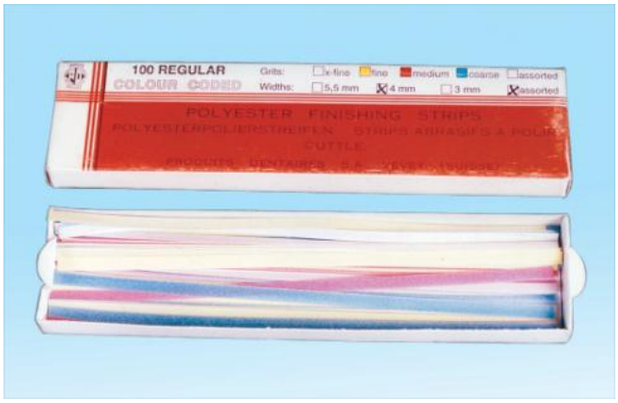


Fig. 23.14A: Interproximal finishing strips



Fig. 23.14B: Composite polishing discs

For the proximal surfaces, the matrix is removed and any excess from the margins is removed using sharp gold knife. The knife is moved from the restoration to the tooth with short, light cutting strokes in order not to disturb the adaptation of resin. The surface and margins of the restoration are smoothened with a cuttlefish finishing strip (strips are used with short strokes in the embrasures).

For a Class V composite restoration, use of sandpaper discs may damage the gingiva and nick the cementum of the tooth. An appropriate finishing bur is used to remove any excess material.

The final lustre is obtained with polishing pastes that may contain pumice, silica, alumina, tin oxide, silicon carbide, zirconium silicate, glycerine, etc. (Fig. 23.15A, B). The commonly used abrasives are pumice and silica bound over cups or strips. White rubber cups are preferred to avoid discoloring the resin. Care should be taken, especially when polishing gingival margin, because the cervical cementum is soft and easily abraded. After the polishing is completed, air is used to blow away any remaining abrasive particles. Dental floss is used to clean the proximal embrasures of any residual paste.

After the polishing, a thin layer of glaze may be applied to improve surface smoothness. Glaze is a film of unfilled polymers with a composition similar to the resin matrix. The life of a glaze is short; get easily abraded.





Fig. 23.15: Polishing paste

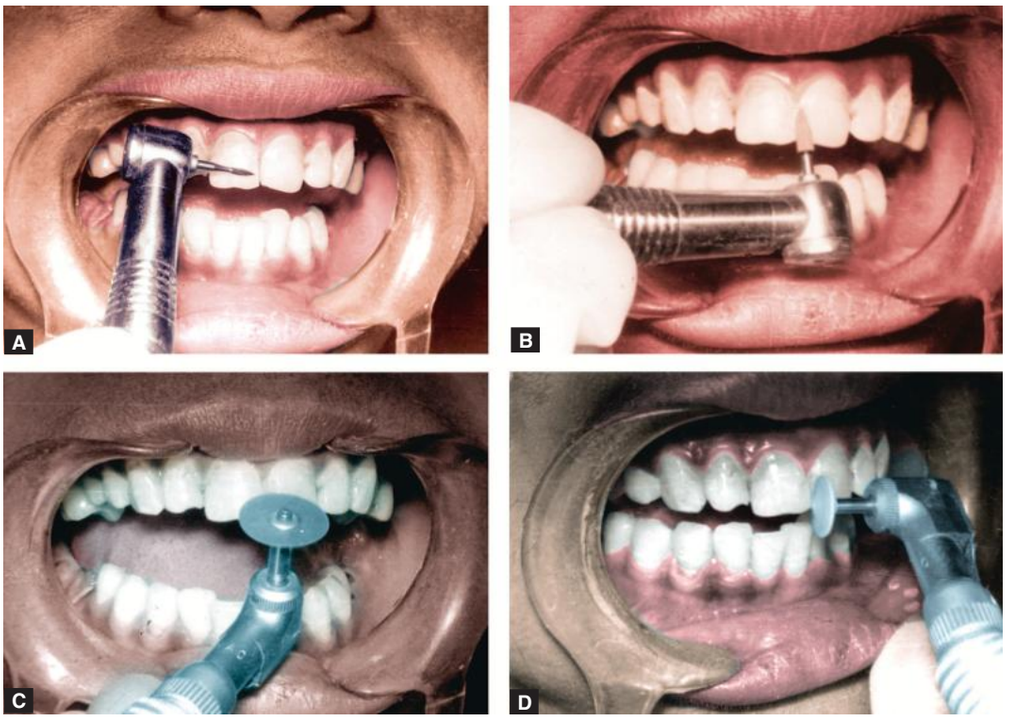


Fig. 23.16: Finishing of composite restoration—maxillary central incisor: **(A)** Gross finishing with tungsten carbide bur; **(B)** Further finishing with silicon carbide stone (green); **(C)** Finer finishing with finishing disc; **(D)** Final finishing with rubber disc

Polishing composites: Instruments used in descending order

- Tungsten carbide bur (12 fluted)/diamond point (30–40 mm) for gross removal of excess material.
- Silicone polishers
- Medium/fine grit finishing discs for proximal areas
- Silicone brushes for polishing pits and fissures
- Polishing pastes using 3.0 mm brushes followed by 1.0 mm brushes
- Applying thin layer of glaze

Polishing agents as surface penetrating sealant have also been used. The finishing process creates microstructural defects or cracks that extend below the surface. Because of extremely low viscosity and wetting ability, the material readily flows across the surface and penetrates into the microstructural defects. Stepwise procedure for finishing of composite restoration of maxillary central incisor is shown in Fig. 23.16A to D; and for mandibular first molar is shown in Fig. 23.17A to G.

The scanning electron microscopic picture of unpolished and polished composite is shown in Figs 23.18 and 23.19.

b. *Indirect Composite Restorations*

The composite inlay/onlay is primarily trimmed and finished on the die, using the same armamentarium as in a direct composite restoration (Fig. 23.20A to D).

At try-in stage, excessive pressure should not be applied while seating (composite material is fragile).

Abrasive discs in a descending order of their abrasiveness are used to adjust the proximal contour and contact (Fig. 23.21). Occlusion should be checked once the restoration has been sealed. The occlusion is adjusted, if need be. The resin cements are usually used as cementing medium. Clear plastic strips secured with wedges are used to prevent flow of excess cement into the gingival embrasures.

After the cementation, the matrix strips and wedges are removed and the margins checked for any excess cement with an explorer tip. Fine grit diamond instruments are used to remove the excess cement. Flame-shaped diamonds are used in the interproximal region, while oval or cylindrical shapes are used occlusally. Finishing with diamond instruments is followed by finishing with 30 fluted carbide burs to obtain a smooth finish. Abrasive strips can also be used in the interproximal region to remove any excess.

C. Glass-Ionomer Restorations

The surface of glass-ionomer cement is sensitive to both dehydration and water contamination during the initial setting phase. The scanning electron microscopic picture of desiccated glass-ionomer cement is shown in Fig. 23.22. If the restoration is

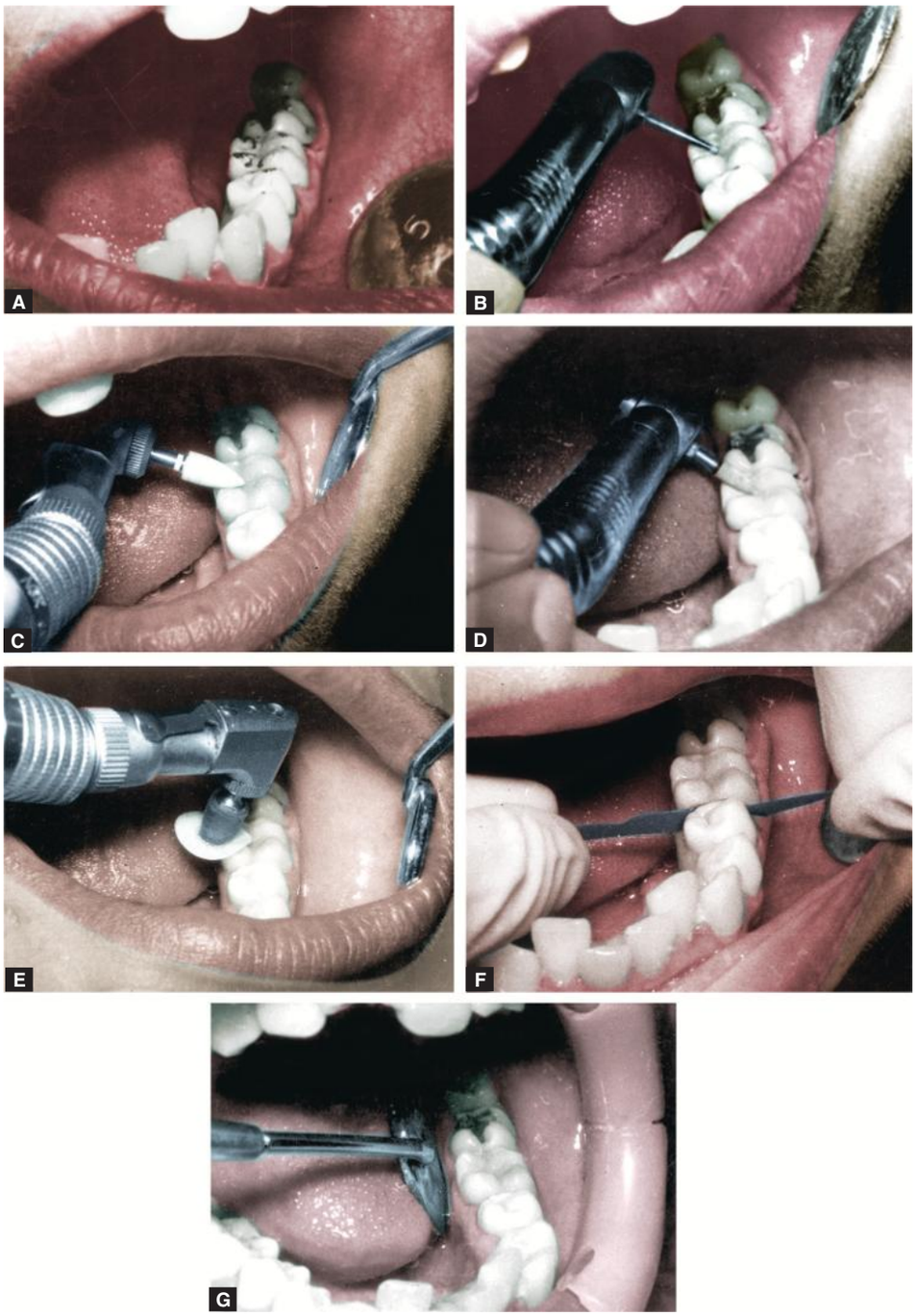


Fig. 23.17: Finishing of composite restoration—mandibular first molar: **(A)** Checking the occlusion of restored posterior tooth with articulating paper; **(B)** Initial finishing with carbide burs; **(C)** Finishing with alpine stone; **(D)** Finishing with fine stone; **(E)** Finishing with disc; **(F)** Interproximal strip used for finishing interproximal area; **(G)** Finished restoration

exposed to air during initial setting, it loses water and develops crazing and cracks. Water contamination on the other hand dissolves the matrix forming ions, resulting in weak and more soluble cement. A damaged restoration stains and dissolves very readily. The finishing and polishing of the restoration is delayed for at least 24 hours, so that the setting reaction is not disturbed (After 24 hours, the surface of the material reaches ionic equilibrium with the oral environment.) In case, light cure glass-ionomer cement is used, the finishing can be started earlier (within 10 minutes).



Fig. 23.18: Scanning electron microscopic view: Unpolished composite surface

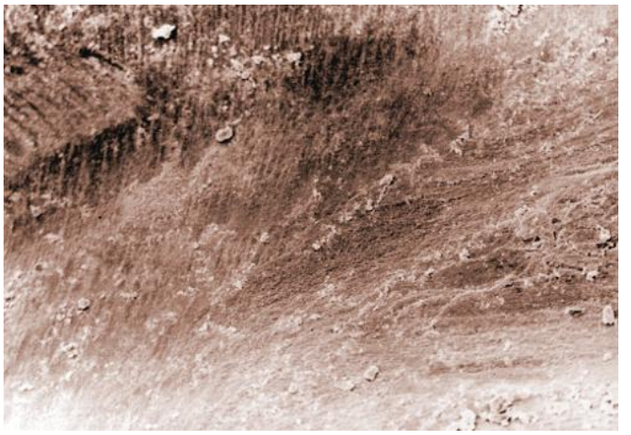


Fig. 23.19: Scanning electron microscopic view: Polished composite surface

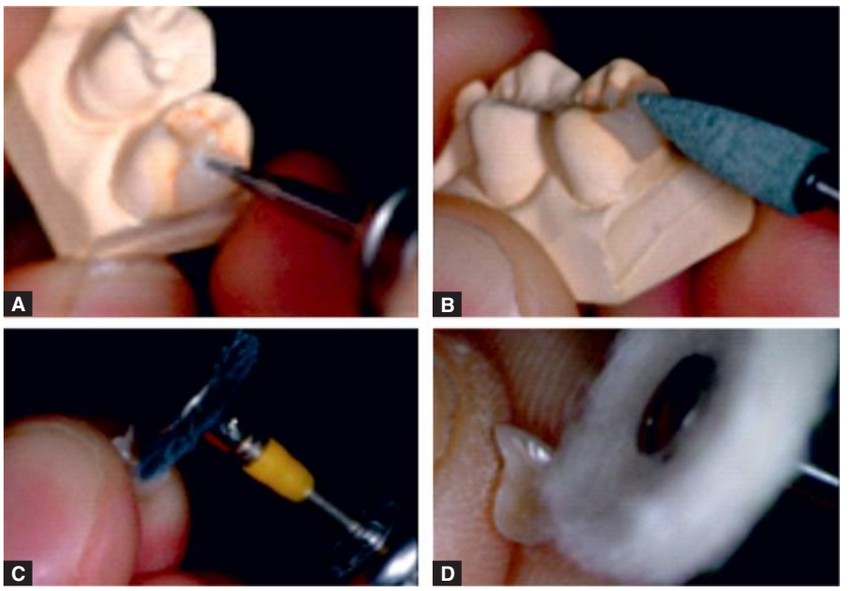


Fig. 23.20: Finishing indirect composite restoration



Fig. 23.21: Proximal finishing and contouring

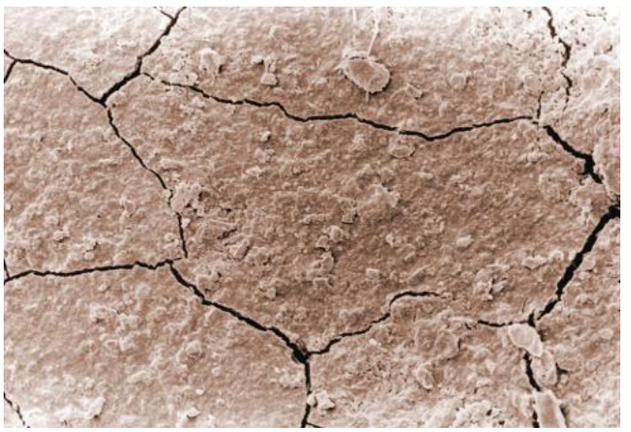


Fig. 23.22: Scanning electron microscopic view: Surface of desiccated glass-ionomer restoration

The ideal surface finish for glass-ionomer cements is produced by the matrix strip. The gross finishing is carried out immediately after the strip matrix is removed. The surface should be coated with a protective layer while the excess is being trimmed. A sharp gold finishing knife is used to remove the excess cement. The trimmed restoration is again coated with cavity varnish/petroleum jelly to protect the cement from moisture. Rotary instruments can also be used at slower speeds to remove excesses at the margins. (Manual cutting instruments may tear the material at the margins, causing marginal breakdown, eventually staining and recurrent caries).

The contours and embrasures are finished with fine diamonds points/12 fluted carbide finishing burs. Margins are finished by using $\frac{1}{2}-\frac{3}{8}$ inch fine cuttlefish discs. Aluminum oxide-coated discs and Soflex discs with Vaseline have shown good results; whereas, use of a tungsten carbide bur disrupted the surface of even mature glass-ionomer cements. A lubricant, such as cocoa butter or petroleum jelly, should be used along with finishing and polishing instruments to avoid excessive heat generation and desiccation. Water spray cooling is not recommended, as it may lead to rough surface. Air coolants may be used. Polyacrylic gel has also been used for polishing glass-ionomer restorations.

The scanning electron microscopic picture of polished glass-ionomer cement is shown in Fig. 23.23.

D. Direct Gold Restorations

Burnishing is the first step in finishing a gold restoration. For the occlusal surfaces, a Spratley burnisher or ball burnisher is moved with considerable pressure from the metal to the tooth surface. A Spratley blade carver is used in the embrasures for proximal restorations.

The optimum contour to the restoration is provided by instruments, viz. Morse scaler, Jones knife or Cleoid discoid carver. The abrasive strips and discs are used to finish proximal areas. Small discs should be used to prevent any damage to the gingival tissue. Abrasives (mostly cuttlefish) used on the discs should be in a descending grade. The discs are rotated from the filling to the tooth surface to achieve smooth margins and surfaces.

Excess gold in the gingival area of the restoration is removed using the contrabevel end of a Wedelstedt chisel. A No. 12 Bard-Parker blade/Wilson's knife may also be used. Discs should not be used in the cervical areas; cementum may get damaged.

After the surface has been smoothened, burnishing is repeated on all the margins. Polishing with tin oxide applied with a rubber cup provides high metallic lustre and satin finish. Polishing instruments should be confined to the restoration surface and used with air coolants.



Fig. 23.23: Scanning electron microscopic view: Surface of polished glass-ionomer restoration

E. Cast Gold Restorations

Initial finishing of a gold inlay is carried out on the die. The casting is cleaned using fine steam at pressure (Fig. 23.24). The casting is examined for any nodules/voids or adherent investment. Defective casting is to be repeated. The casting is pickled in 50% hydrochloric acid and neutralized in sodium bicarbonate. The clean casting is tried in the die. Once satisfied with the accuracy of the casting, the sprue is separated using a separating disc. The stub of the sprue is finished according to the contour required.

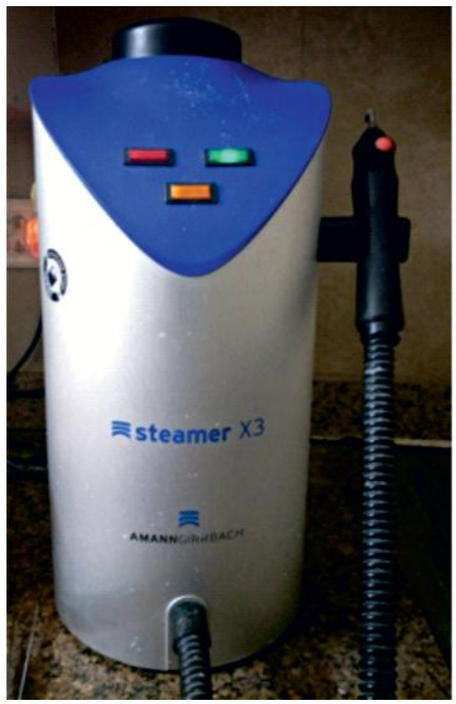


Fig. 23.24: Steam cleaner

Procedure

Any extension of the casting beyond the margins is removed with a small cylindrical diamond point rotating away from the operator. Occlusal surface is finished using abrasive stones and discs in an order of descending abrasiveness (Fig. 23.25).

The proximal surface is contoured lightly with a ½"–⅝" carborundum disc followed by smoothening with medium and fine cuttle paper discs.

The inlay/onlay can be tried in the cavity after proper isolation. A throat screen (3 × 3 gauze sponge) is placed over the posterior of tongue to prevent any accidental ingestion of the casting. The inlay is seated into the prepared cavity with mild pressure. If the casting does not seat completely, it can be because of heavy proximal contacts, uneven internal surface or an actual distortion. A dental floss passed through the contact area gives an idea of tightness of contact and should be accordingly reduced.

After the casting is seated into the cavity, the occlusion is checked with an articulating paper. The occlusal contacts are first verified in the intercuspal position followed by lateral working and non-working positions and finally the protrusive and retrusive positions.

The next step is to check the marginal relationships under a magnifying glass by running a probe across the metal–tooth interface. In cases, where the margin of the inlay is bulky or override the unprepared tooth surfaces, the excess is reduced with a carborundum disc or fine cuttle discs.

Polishing is carried out with a No. 11 Robinson bristle brush and Tripoli. The brush should always be moved from the casting to the tooth surface. For achieving a very high lustre, a chamois wheel with rouge may be used (Fig. 23.26A, B). Over polishing should be avoided as it may ruin the anatomical contours or break the thin margins leading to an open casting.

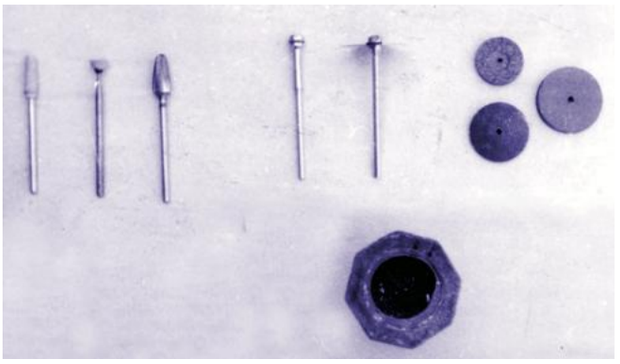


Fig. 23.25: Finishing instruments for cast gold restoration



Fig. 23.26A: Polishing kit for cast gold restoration

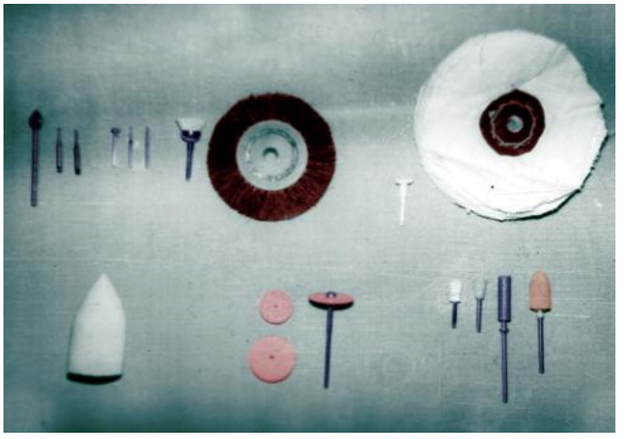


Fig. 23.26B: Polishing cast gold restoration with brushes and buffs

F. Non-Precious Alloy Restorations

Nickel–chromium is commonly used non-precious alloy. Chromium provides high resistance to corrosion and quickly forms an 'inert' or 'passive' oxide layer on the surface; thereby protects underlying metal from the oral environment. However, because of the increased hardness of these alloys compared to the gold alloys, special equipment is required to trim and polish these restorations.

The instrument kit required for finishing non-precious alloy is shown in Fig. 23.27A. Hard abrasives are used to finish non-precious alloys in a sealed chamber (Fig. 23.27B).

The casting is separated from the investment and cleaned under steam to remove adherent investment material and the oxide film. It is advisable to achieve the requisite shape and contours on the wax pattern, so as to minimize any trimming of the casting.

Electrolytic polishing is commonly used as a part of the finishing process for non-precious alloys, as it reduces the time and effort for mechanical finishing.



Fig. 23.27A: Instrument kit for finishing non-precious alloys



Fig. 23.27B: Finishing non-precious alloys

Electrolytic polishing works on the principle of reverse electroplating in which the restoration acts as the anode. The electrolyte is usually a mixture of sulphuric acid, phosphoric acid, glycerine and water. A current density of 0.5–1.0 amps/mm^2 at voltage of 4–10 V is passed for a few minutes at room temperature. The anode, i.e. the restoration ionizes and is depleted of a very small amount of alloy (a few angstroms) more readily on the rough surfaces than the smooth surfaces. These first products of electrolysis are collected in the hollows of the rough metal surface. The prominences are gradually dissolved and a smooth surface results.

G. Porcelain Restorations

There are three methods of fabrication of ceramic inlays and onlays:

a. Conventional method using firing porcelain,
b. Milling ceramic blocks using machinable ceramic
c. Casting by the lost wax technique using castable and pressable ceramic.

The second and third methods do not use the application of a glaze layer to attain the final surface finish.

The technique for finishing and polishing of the restoration by each method is as follows.

a. *Restoration formed by Firing Porcelain*

The desired finish on the fired ceramic restoration is achieved by glazing and to a smaller extent by polishing. A glazed surface of porcelain is comparatively much stronger than the unglazed porcelain.

Positive effects of glazing:

- Glazed surface leads to reduced wear of opposing surfaces.
- Oral functions are improved; food slides easily over the restoration.
- Improved oral hygiene (Rough/grinded/unglazed surface invites bacterial accumulation.)
- Restorations are esthetically pleasing (Light reflection is appropriate.)
- Glazing provides proper strength (Strength is reduced during grinding/adjusting ceramic restoration.)
- Glazing reduces flaws and eliminates crack propagation.

Once the glaze layer is removed, transverse strength of the restoration almost reduces to one-half. This situation has a clinical significance, since the restoration often requires occlusal adjustment after placement. Any adjustment makes the surface rough and may lead to loss of glaze layer, thereby reducing the strength of the restoration. However, with the use of polishing systems, like Soflex discs, Porcelain polishing kit, etc., it is possible to minimize the damage caused by intraoral adjustment.

Types of Glazing

Glazing is of two types:

i. **'*Autoglazing*':** It is a process in which a smooth surface is obtained without the use of an additional glaze. By careful control of the furnace temperature (955°C–1065°C), a glaze is created. The surface porcelain melts slightly, flows and glazes; may result in little roundening of the corners. The disadvantages with this technique are that any excessive glazing would destroy the shape of porcelain, increase porosity, reduce strength and may alter the color. This technique requires great precision, but provides adequate esthetic appearance.

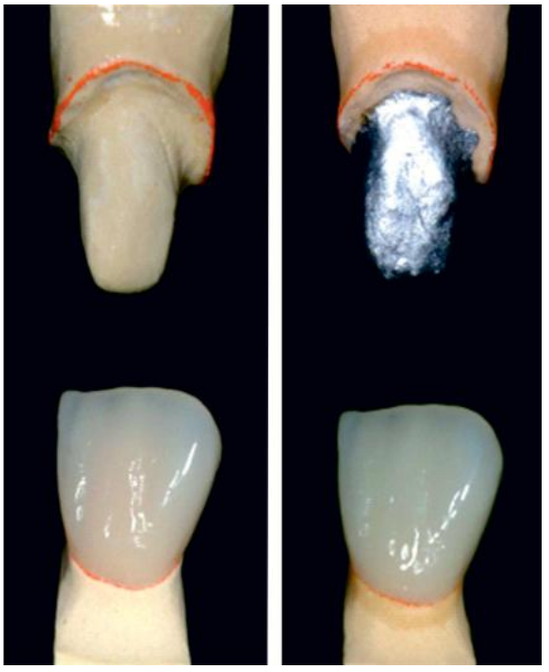


Fig. 23.28: Glazed ceramic restoration

ii. *'Overglazing':* It is the application of an external glaze layer over the porcelain surface. Such a glaze contains silica and low fusing fluxes and increases translucency (Fig. 23.28).

The fabricated ceramic restoration is separated from the die and cleaned of any adherent investment with a coarse brush. Metallic instruments should be avoided, as they tend to leave gray marks on the restoration. The restoration is seated on the master die for adjustments and finishing. Care should be exercised during handling, as ceramic is very brittle and susceptible to fracture.

The finishing methods are similar to those described earlier for an indirect composite restoration. During the process, care should be exercised to preserve the glazed surface as much as possible. After adjustments, the areas where the glaze has been damaged, is repolished.

Finishing ceramic restoration in sequence

- Use of fine grit diamonds.
- 30 fluted carbide finishing burs.
- Rubber abrasive points and cups at slow speed with air water spray.
- Diamond polishing pastes applied with a bristle brush.

Before the ceramic inlay/onlay is cemented, the cavity side of the restoration is etched. The etched ceramic surface is treated with a silanating agent to improve its bond with the cementing medium. A dual cure composite is preferred for luting so as to ensure complete polymerization, even in areas where light may not reach.

b. *Restorations formed by Milling Ceramic Blocks*

The CEREC system utilizes CAD-CAM technology, and CELAY system utilizes precise milling machine.

The CELAY system uses a 'proinlay' made of composite as a template. The proinlay may be fabricated either directly in the oral cavity or indirectly in the laboratory.

Checking for any seating interferences and correcting the occlusion are also carried out on the die, thereby reducing the chairside finishing and polishing time.

The milled restoration is adjusted using coarse to fine diamond instruments with air–water spray as coolant. Since these systems mill the ceramics, a rough surface is produced on the casting initially. The rough surfaces are polished to a high gloss using Soflex discs.

The final occlusal adjustments, if needed, are carried out with rotary diamonds followed by 30 fluted carbide finishing burs. Areas of adjustment are repolished using the same procedure, i.e. flexible polishing discs and diamond polishing pastes.

Methods of polishing of adjusted ceramics

- Sandblasting with aluminium oxide before refiring.
- Series of Soflex discs designed for finishing of composite.
- Instrument which progressively decreases in particle size; 30 fluted carbide bur and diamond polishing paste.
- Fine diamond burs followed by diamond paste.
- Diamond paste and aluminium oxide paste for final polishing.

Any excess composite in the proximal embrasure is removed with finishing diamonds and polishing strips.

c. *Restorations formed by Castable Ceramic*

The castable ceramic restoration is finished and polished on the master die, as described earlier for other ceramic restorations (Figs 23.29A, B and 23.30A to D).

After the occlusal adjustments, the colorant shades are applied on the exterior surfaces and baked. The disadvantage with castable ceramic is the loss of external stains and esthetics. The restoration is cemented into the cavity and chairside finishing and polishing is carried out as described before.

Different systems used for finishing and polishing are summarized in Tables 23.1 and 23.2.

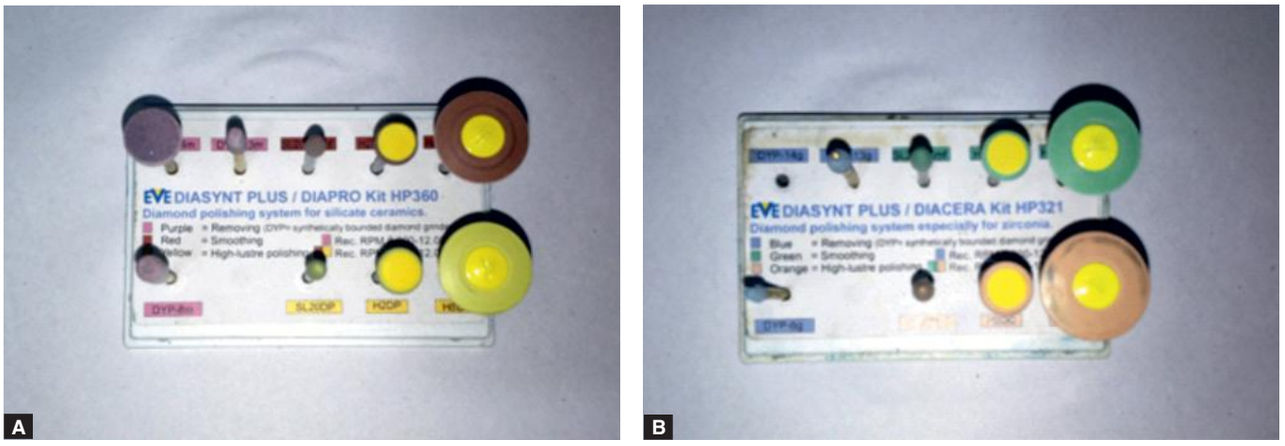


Fig. 23.29: Finishing kits: **(A)** For conventional ceramic; **(B)** For zirconium

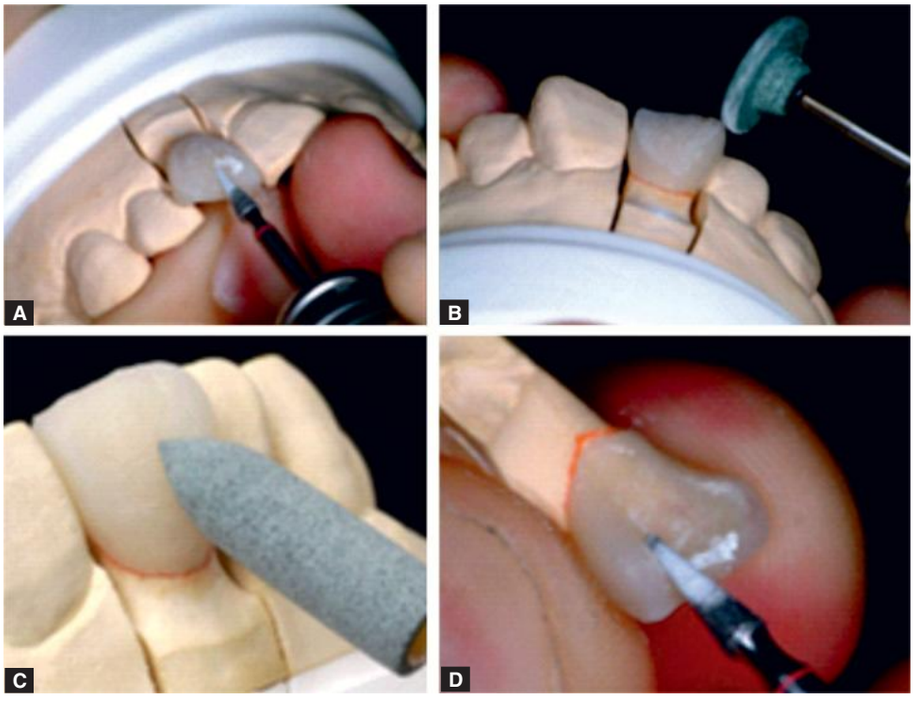


Fig. 23.30: Finishing of ceramic restoration

Concept of Finishing Polish

A 'finishing polish' is considered as the last step in polishing (can be designated as final polishing). Finishing polish removes residual oxidation layer and left-over minor imperfections. The modality provides smooth polish without any surface cutting. A few agents contain oil in polishing pastes to enhance glaze. A polishing pad is used with gentle finger pressure to achieve the requisite finishing polish. A few authors have labeled the technique as 'Jeweling'.

HAZARDS OF FINISHING AND POLISHING

The hazards involved with finishing/polishing procedures are:

a. *Production of aerosol:* An aerosol is a dispersion of fine solid or liquid particles (e.g. tooth structure, restorative material or micro-organisms) in air. Aerosols may remain suspended in the air for even more than 24 hours, thereby promoting cross-contamination. A major pulmonary disease that may occur with the chronic exposure to aerosols

Table 23.1: Different systems used for finishing and polishing

Abrasive used	*Trade name*	*Manufacturer*
Aluminum oxide	Soflex XT discs • Coarse grit • Medium grit • Fine grit • Superfine grit	3M Health Care
Aluminum oxide	Shofu super snap disks • Medium grit • Fine grit • Ultrafine grit	Shofu Dental Co.
Silicon carbide	Moore's X and XX (Waterproof) disks	E.C. Moore Co.
Rubber wheels and points	Composite and quazite	Shofu Dental Co.
Alumina pastes	Enhance finishing and polishing strips	Dentsply
	Neos system	Dental Schweiz
Rubber abrasive points and cups	Porcelain polishing kit	Shofu Dental Co.
Diamond polishing paste	Porcelain polishing system	Brassler

containing silica is the 'silicosis' or 'Grinder's disease'. Silica is a very common component of restorative and polishing materials. Aerosol hazards can be minimized by following procedures:

- Adequate ventilation in the operatory that filters harmful particles from the air
- Use of safety glasses with side shields, disposable face masks and gowns
- Use of rubber dam protects the patient against oral inhalation of aerosols
- Use high vacuum suction.

b. ***Production of vapors:*** High temperature generated during finishing of amalgam may form mercury vapor. Thermal decomposition of polymeric materials, like acrylic resins and composites, produces monomer vapors. The production of vapors can be minimized by using low speeds and intermittent cutting. If using high speeds, copious air-water spray should be used as a coolant. Inhalation of these vapors can be prevented following the same procedures as given for aerosols.

c. ***Pulpal reactions:*** The finishing and polishing instruments may raise the temperature by at least 20°C. If the remaining dentin and enamel thickness is less, this heat is rapidly conducted to the underlying pulp resulting in various pulpal ailments. Proper use of coolants, such as water, air and air-water spray protects against excessive rise in temperature. Cup brushes are preferred for polishing as compared to rubber appliances as the former generate less heat. Also, use of intermittent polishing at low speeds is preferred to prevent excessive heat production.

d. ***Soft tissue injuries:*** Soft tissues, like lips, cheeks, tongue and gingiva may be injured, if adequate care is not taken to protect them.

e. ***Miscellaneous***

- Impact of the instrument may damage the filler particles of composites.
- Voids incorporated during the filling process may be exposed, lodging unsightly strains.
- Composites can be lodged in the oral soft tissues during finishing and shaping procedures, which can lead to persistent chronic inflammation of those tissues.
- Beryllium fumes could be formed when finishing chrome–cobalt alloys containing beryllium. Beryllium is also a known carcinogenic.
- Nickle, which is also a component of chrome–cobalt alloys, has an ability to produce allergy.

The well-finished and polished restorations are not only esthetically pleasing, but are also conducive to the oral environment. The proper marginal adaptability of the restoration saves an individual from periodontal problems, subsequently achieving better clinical success of the treatment.

Table 23.2: Ceramic restorations: Finishing and polishing systems

Finishing/polishing agents	*Ingredients and action*
DiaSheen polishing paste	Polishing paste in fine, medium and coarse grits used with stiff bristles and felt wheels (range from gross trimming to fine high glass polish)
OptraFine	Impregnated rubber discs for chairside polishing
CeraMaster	Blend of silicone and diamond particles (help finishing from coarse to fine surface)
Diamond polishing paste	Diamond polishing paste, used with buff disc for final polishing.
Diashine polish/brush	Diamond grit polish used with soft brushes for final polishing.
Brite polish	For final polishing

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